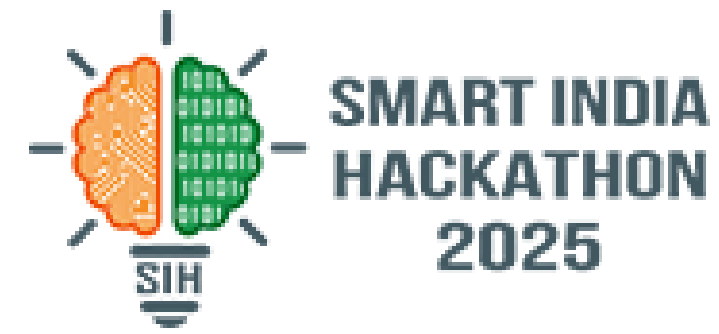


SMART INDIA HACKATHON 2025



- **Problem Statement ID –25159**
- **Problem Statement Title- Real-Time AI/ML-Based Phishing Detection and Prevention System.**
- **Theme-Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID-**
- **Team Name:Visioncraft.**



Real-Time AI/ML-Based Phishing Detection and Prevention

Proposed Solution (Describe your Idea/Solution/Prototype)

It uses **multi-modal (NLP + Visual + Graph Models)** to detect and block phishing attempts across email, SMS, IM, and websites in real time (<100 ms).

Detects **zero-day attacks** (new domains, AI-generated content).
Blocks **URL redirections, cloaked links, homoglyph domains**.
Prevents engagement by users before the attack succeeds.

Combines **Transformer-based AI scoring + sandbox behavioral analysis + domain reputation** matching through the Supabase malicious database to achieve high-accuracy detection.

Explainable AI shows exactly **why a link was flagged as phishing**, combining our model's reasoning with **VirusTotal's multi-engine** threat verification.

Lightweight browser & WhatsApp extensions deliver these insights in real time without slowing user activity.



TECHNICAL APPROACH

Techstack Used:

- 1.React Framework
- 2.Python(FastAPI)
- 3.XGBoost
- 4.Transformers(Vision + Text)
- 5.Chart.js
- 6.Scikit-learn
- 7.Supabase
(Auth + Databases + API)
- 8.Sandbox API

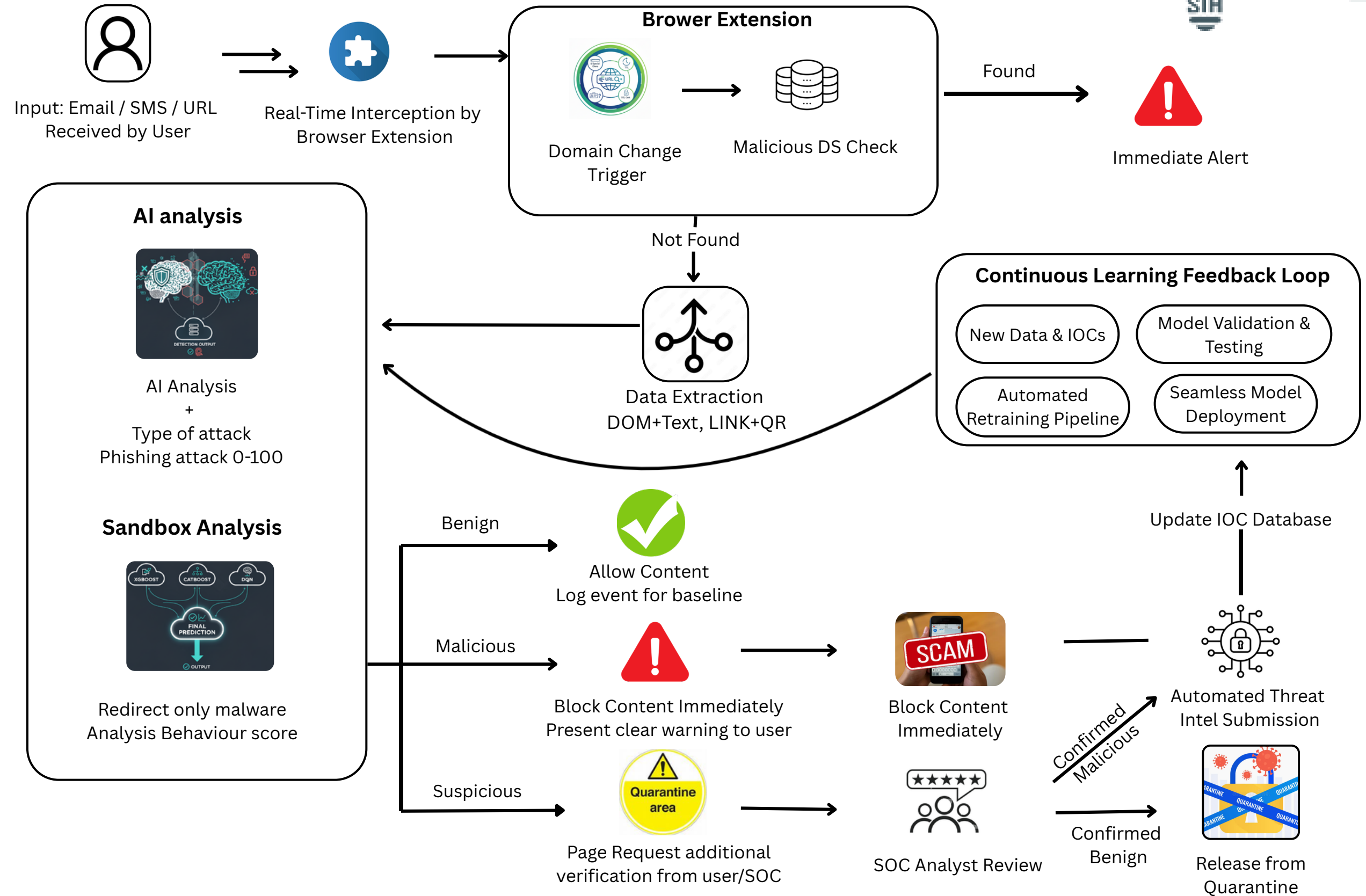


Prototype:

Link [Quorix](#)

Prototype Video Link:

 [Youtube Video](#)



Analysis of the feasibility of the idea	Potential challenges and risks	Strategies for overcoming these challenges
Technical – Achievable	Latency& real-time performance	Use AI, optimized for ultra-fast (<50 ms) edge-inference. Lightweight browser extension ensures minimal overhead.
Accuracy – Effective	Evasion, cloaked redirects, zero-day attacks	Combine AI scoring + Sandbox behavior analysis + malicious-DB lookup for multi-layer detection. IOC DB updates continuously improve threat coverage.
Data – Enrichable	Compliance	Automated collection of malicious events is stored in Supabase and used to update the IOC database .

Given that feasibility and viability being our top priority we will ensure to keep everything

- **Technically viable** – Uses proven tools (AI, Supabase, VirusTotal), no heavy on-device ML.
- **Economically feasible** – Low-cost cloud setup, scalable for enterprises & institutions.
- **Operationally practical** – Simple browser extension, auto-detection, user alerts, admin dashboard.

IMPACT AND BENEFITS

Potential Impact on Target Audience

1. Enhanced Security for Organizations

- Prevents financial losses from BEC and ransomware.
- Protects brand reputation and ensures compliance.

2. Empowerment for SMBs

- Provides enterprise-grade security affordably.
- Safeguards business continuity against cyber threats.

3. Protection for Individual Users

- Shields personal data and financial accounts from theft.
- Builds digital literacy through real-time warnings.

4. Efficiency for IT Teams

- Reduces alert fatigue with high-accuracy AI.
- Automates threat triage for faster response.

5. Resilience for Government

- Secures critical infrastructure from attacks.
- Maintains trust in essential public services.

Benefits of the solution

1. Social Benefits

- Builds a safer, more trustworthy digital environment.
- Protects vulnerable groups like the elderly.

2. Economic Benefits

- Saves billions lost globally to cybercrime annually.
- Prevents costly operational downtime for businesses.

3. Environmental Benefits

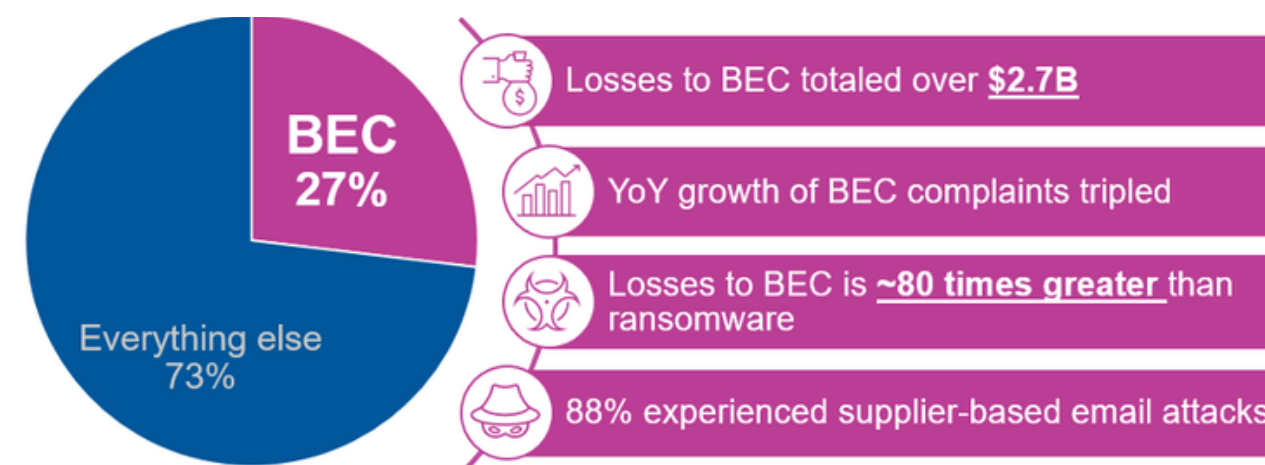
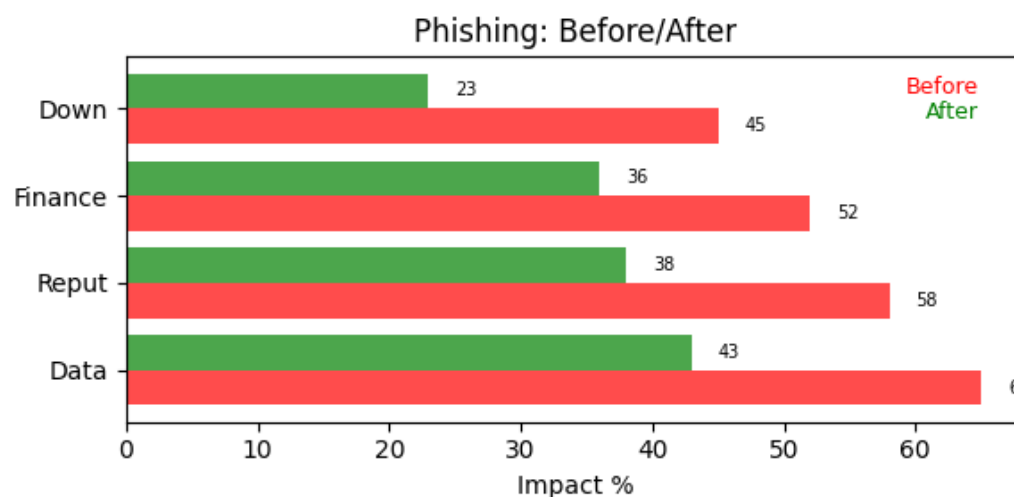
- Uses energy-efficient AI models.
- Reduces e-waste by preventing hardware-compromising attacks.

4. Technological Benefits

- Makes advanced AI security accessible to all.
- Drives innovation in cyber defense techniques.

5. Strategic Benefits

- Supports national cybersecurity initiatives.
- Enhances collective defense through threat intelligence sharing.



**Make the data safe and
have latency < 100ms**



RESEARCH AND REFERENCES



Research Survey

Aspect	Conventional Systems	Proposed System
Detection	Blacklists & static rules	AI + threat intel, zero-day ready
Features	Manual feature engineering	Auto feature learning (DL + NLP)
Security	Vulnerable to adversarial tricks	Adversarial training + graph defense
Response	Reactive alerts only	Proactive block + auto response
Learning	Manual retraining needed	Continuous self-learning pipeline

Reference

Research Paper:

[1] Abdul Rafay, Naveed Akhtar, Muhammad Farhan, “Phishing Attack: Its Detections and Prevention Techniques” [Online].
Available: https://www.researchgate.net/publication/374848676_Phishing_Attack_Its_Detections_and_Prevention_Techniques (Oct 2023).

[2] Nikhil Pandey, Harshal Patil, Mayur Patil, “Next Generation Phishing Detection and Prevention System using Machine Learning” [Online].
Available: https://www.researchgate.net/publication/369784414_Next_Generation_Phishing_Detection_and_Prevention_System_using_Machine_Learning (Apr 2023).

[3] Jaya B. Nair, Nithin Mathew, Rohit R., “AI-Based Phishing Detection and Prevention System Using Deep Learning and NLP” [Online].
Available: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10735206> (Sep 2024).

Dataset:

[1][Website & URL Analysis](#) [2][Textual & Persuasion Analysis \(NLP Layer\)](#) [3][Real-Time Phishing URL Feed](#)